

Lesson: Control Systems and Units

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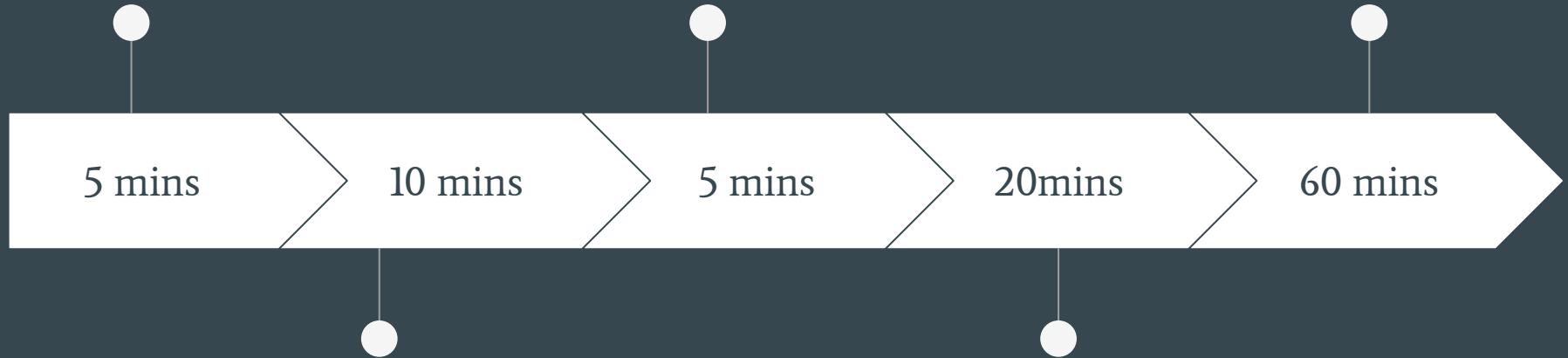
Objective

- Cover basics of control theory
- Cover basics of units and importance of them
- Cover basics of motor control using P/PD/PID control

What is a “plant”

What are common FRC
units?

Interactive
Demo/Assignment



What is a control
system

Putting it all together

What is a “Plant”

- “A plant in control theory is the combination of process and actuator.” - Wikipedia Plant (Control Theory)
- In Robotics, there are many types of plants:
 - The robot itself
 - A motor
 - A module
 - A subsystem
 - Etc.
- How do we know how to treat each plant?
 - Math!

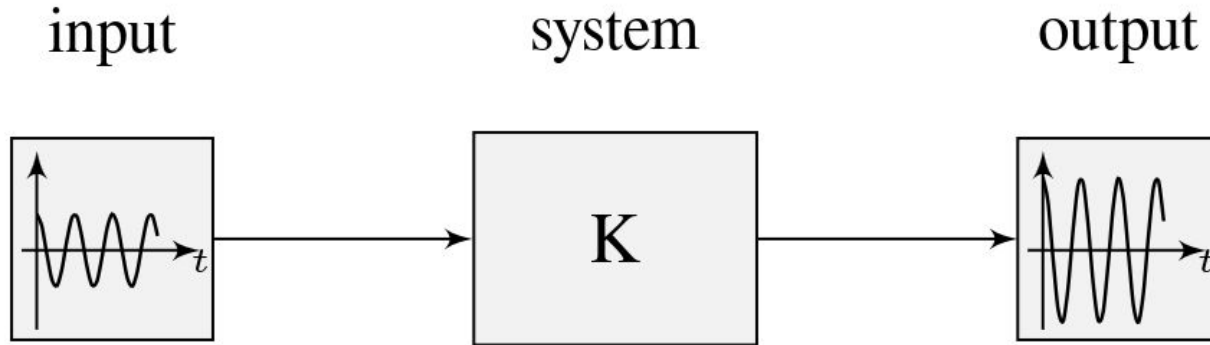
What does a DC Motor look like in Math terms?

- Relationship between position (Theta) and Voltage in the LaPlace transform
- This is called the “Transfer function”

$$\frac{\theta(s)}{V(s)} = \frac{K_t}{s [(Ls + R)(Js + b) + K_t K_b]}$$

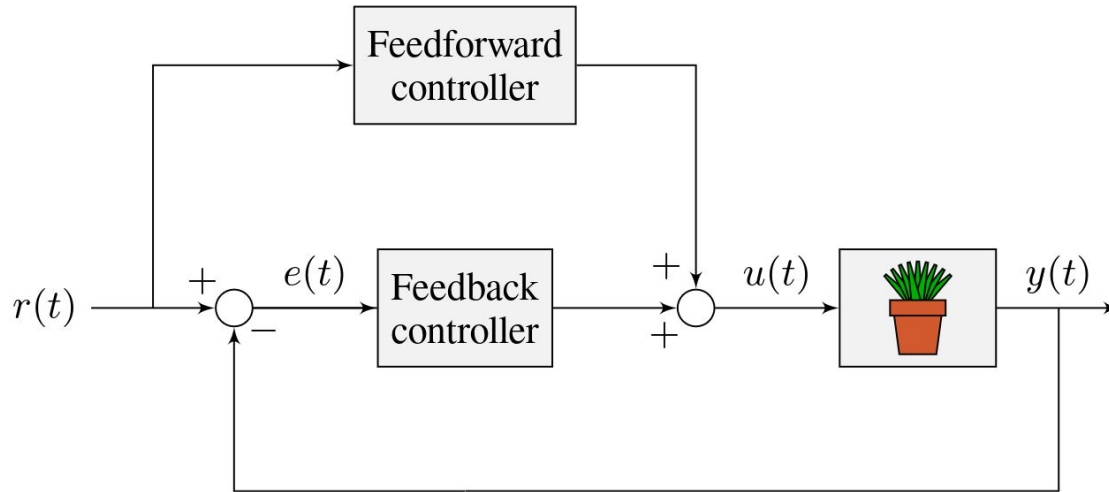
What is Gain?

- Gain is the change applied to the input signal
- Notation is to use “K” and subscripts



What are feedforward and feedback controllers?

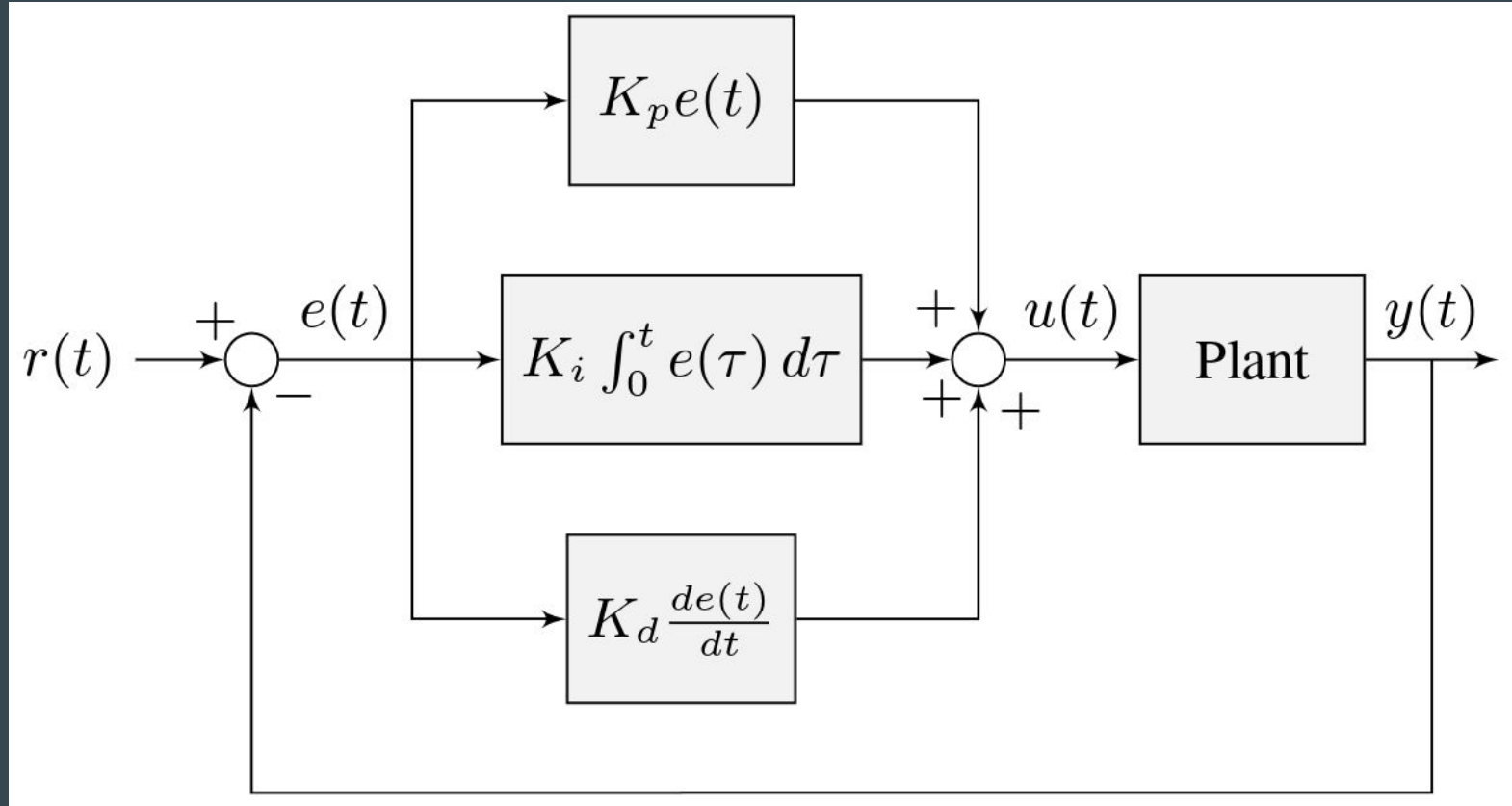
- Feedforward applies a Gain to the input and directly links it to the output
- Feedback applies a Gain to the input determined by the information the Plant provides



What control systems do FRC robots use?

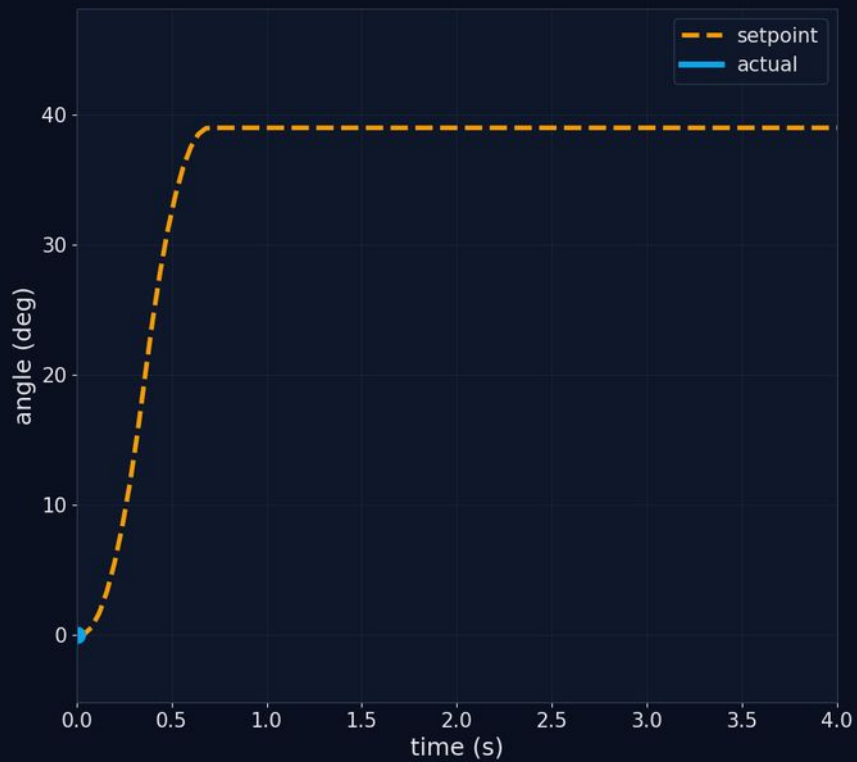
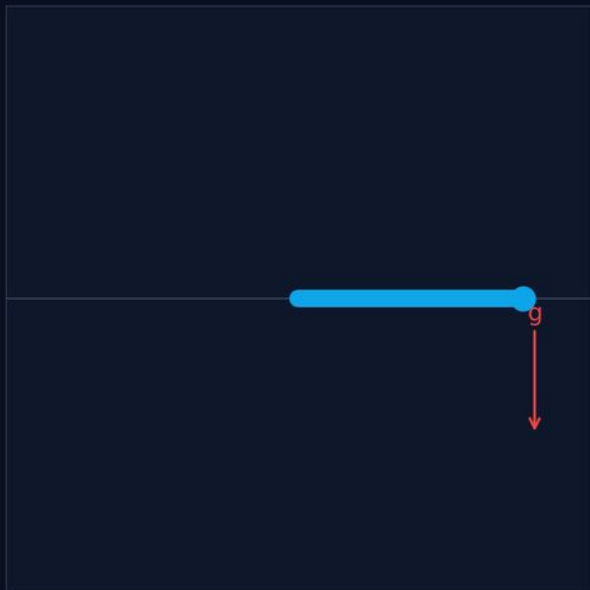
- PID is common in FRC however, most are really just PD
- What do these Gain terms mean?
 - K_p is Proportional Gain, applying a linear gain based on the error
 - K_i is Integral Gain, work on accumulated gain
 - K_d is Derivative Gain, working on predicted future error
- Why only PD?
 - Most FRC components being motor position control don't need Integral control because of the transfer function.

PID Control Block Diagram

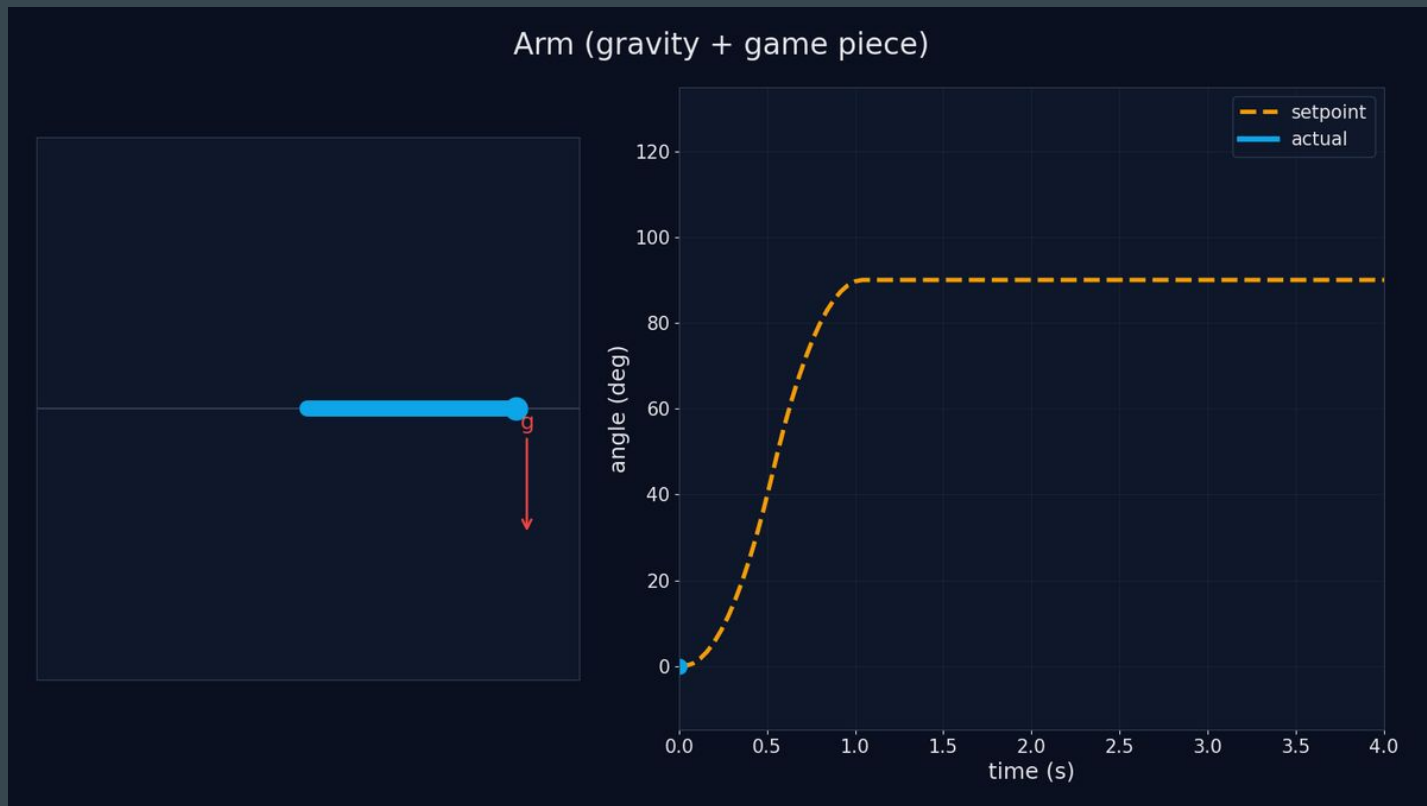


Examples!

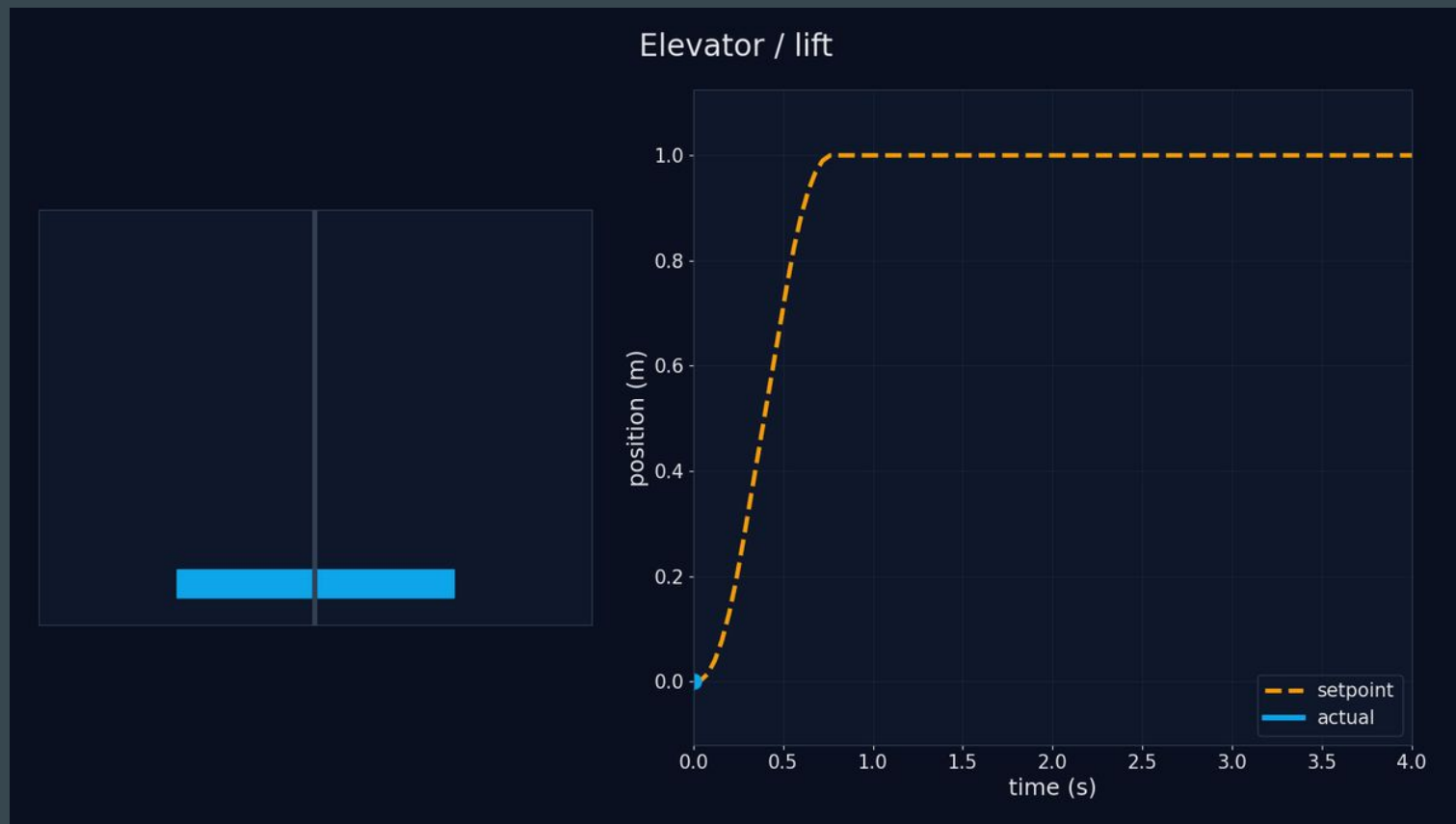
Arm (gravity + game piece)



Examples!



Examples!



Units!

- Units are very important to FRC
- Most of FRC is designed mechanical in SAE Units (Inch, lbs, etc)
- However, software almost always expects Metric Units. This is partly because Metric Units can take advantage of redux math to solve division problems quickly.
- Always, always, always check units.

Examples:

- FRC Robot rules limit weight to 115lbs however, the drive constraints need to know the robot weight in Kg.

Homework

Tune each of the FRC control system examples on the website:

<http://18.118.112.195:8000>

References

Control Systems for FIRST Robotics Competition:

<https://file.tavsys.net/control/controls-engineering-in-frc.pdf> (This is a great resource!!!)